



Experience

Senior AI Engineer

Coforge Limited, Pune

- Designing and developing enterprise-grade Agentic AI systems using Supervisor LLM orchestration patterns, where complex workflows are decomposed into specialised callable agents with defined input/output contracts, error recovery strategies, and stateless tool interfaces.
- Building secure, multi-agent pipelines for financial and resource intelligence use cases, integrating LLM-driven decision making with deterministic rule layers to ensure reliability and auditability in production environments.
- Developing Gen AI applications end-to-end — from requirement gathering and solution architecture through to deployment — covering LLM integration, prompt design, agent tool specification, and cross-agent handoff contract design.
- Architecting ML forecasting systems combining gradient boosted models (LightGBM, XGBoost) with quantile regression and SHAP-based explainability, deployed within agentic pipelines where ML agents are orchestrated alongside ingestion, feature engineering, and reporting agents.
- Implementing data security controls within AI applications, including IT-governed access management and role-based deployment to ensure sensitive enterprise data is protected across the entire agentic workflow.
- Driving end-to-end delivery of AI solutions: from data pipeline design and feature engineering through model training, agent orchestration, dashboard integration, and stakeholder-ready reporting.

Senior Machine Learning Engineer

Optisol Business Solutions , Coimbatore

- **Leading the Computer Vision and ML team**, focusing on building scalable AI solutions for industrial use cases in **agriculture**, construction, and manufacturing.
- **Driving end-to-end project delivery**: requirement gathering, solution architecture, POC development, deployment, and optimization.
- Developed and deployed models for **object detection, segmentation, and classification** using **YOLOv5/v8, SSD, Detectron2**, and **custom CNNs**.
- Integrated **AWS Machine Learning services** such as **SageMaker, Lambda, S3, and Step Functions** to streamline the ML lifecycle — including **data preprocessing, model training, tuning, monitoring, and deployment**.
- Implemented **CI/CD pipelines** for model training and deployment using **SageMaker Pipelines and CodePipeline**.
- Fine-tuned models on domain-specific data to improve accuracy and reduce false positives, ensuring models meet real-world reliability standards.
- Mentored junior engineers and coordinated with clients for technical discussions and solution validation.

Machine Learning Engineer

Optisol Business Solutions , Coimbatore

- Built and deployed Full Stack ML solutions using **TensorFlow, PyTorch, OpenCV, and CUDA** for real-time object detection, segmentation, and measurement systems.
- Designed **vision-based automation** for industrial clients, with focus on **edge deployment** using **Docker** and lightweight model architectures.
- Developed REST APIs using **Flask** to integrate ML models into production environments.
- Promoted to **Team Lead** for delivering successful POCs and managing end-to-end project workflows.
- Collaborated closely with cross-functional teams and clients to translate business requirements into robust AI solutions.

Projects

Project Title: Project Resource Intelligence Platform

Description:

- An end-to-end ML-based cost forecasting and budget risk prediction system built for project resource management, operating exclusively on CAPITAL cost data.
- The system predicts how much cost a resource will utilise on a project for each remaining fiscal month, and proactively identifies whether a project is at risk of going over or under budget — providing project managers 2–3 months of lead time for corrective action.
- Built as a secure, installable enterprise application accessible only to authorised managers with IT-governed permissions, ensuring sensitive financial and resource data is protected end-to-end.

Achievements:

- Designed and implemented a **5-agent agentic pipeline** orchestrated by a Supervisor LLM, where each agent is exposed as a callable tool with defined input/output contracts, error codes, and recovery strategies.
- Built the **Ingestion Agent** — a 6-stage Python pipeline that loads, validates, domain-splits, and persists raw project cost data into clean partitioned datasets, with upsert logic, correction detection, and full audit trails.
- Built the **Feature Engineering Agent** — engineered 25 features across 5 groups (Historical Cost Behaviour, Calendar & Working Hours, Project & Budget Context, Resource Context, Temporal), implementing a 3-lag rule engine to prevent data leakage between training and scoring datasets.
- Built the **ML Agent** — Model A, a quantile regression ensemble (LightGBM + XGBoost + Linear Regression) producing P10/P50/P90 cost forecasts per resource-project-month, and Model B, a gradient boosted classifier predicting project budget risk labels (OVER_BUDGET / ON_TRACK / UNDER_BUDGET) with a deterministic CPI-based rule override and SHAP explainability.
- Designed cross-agent **manifest-based handoff contracts** enabling stateless, independently retryable, and testable tool invocations across the entire pipeline.
- Packaged the entire system as a **secure enterprise application**, restricted to authorised managers only, installed exclusively via IT department approval to enforce data governance on sensitive financial and resource data.

Technologies Used: Python, LightGBM, XGBoost, scikit-learn, SHAP, pandas, FastAPI, MLflow, Apache Airflow, Feature Store, Power BI

Project Title: Cogo Insurance

Description:

- Cogo Insurance is a Usage Based Insurance provider in the USA predominantly catering to trucks and heavy vehicles.
- Usage-based Insurance schemes aim to reward drivers with safe driving habits by charging them with less premiums and penalizing drivers with bad driving habits by charging them more premiums.
- Our objective is to identify driver's behavior based on telematics data and assign insurance premium based on that driver's behavior.

Achievements:

- Derived features such as hard acceleration, hard braking and hard cornerings to identify driving behavior.
- Used 3rd party API's such as WeatherAPI to determine if a driver is driving in harsh weather conditions.
- Built a driver score calculation formula based on weighted average method using the derived features.
- Built and deployed boosting based supervised machine learning models to detect the probability of driver getting involved in accidents.
- Built and deployed anomaly detection models to detect abnormal vehicle behavior.
- Setup a pipeline using AWS Glue to process the daily telematics data for around 20000 drivers and automatically calculate the driver score for those drivers and these driver score would be stored in AWS RDS MySQL database.
- Built a FastAPI to fetch the driver scores based on driver ids from the AWS RDS database.

Technologies Used: AWS Glue, AWS Databricks, AWS RDS, AWS S3, Fast API

Project Title: Utility Pole Project

Role: Project Lead

Description:

- Led a team of 12 members in a comprehensive utility pole project involving instance segmentation using MMDetection and OpenCV.
- Created a 3D pole dataset and developed a JSON file parser for data handling.
- Implemented various features including height prediction, super-resolution image generation, and blocky node creation.

- Designed and developed Python code following Object-Oriented Programming principles for seamless integration and maintenance.
- Maintained regular communication with clients, ensuring project requirements were met on time.
 - Ensured project success by overseeing the delivery of milestones and updates according to schedule.
- Technologies Used:**
- MMDetection ,OpenCV, Python, JSON
- Achievements:**
- Successfully delivered a complex utility pole project meeting client specifications and deadlines.
 - Enhanced team productivity through effective leadership and coordination.
 - Implemented innovative features improving the accuracy and efficiency of instance segmentation and data processing tasks.

Skills

Technical: Machine Learning, Deep Learning, Python, Natural Language Processing, CNN, RNN, Encoder - Decoder models, Open-CV, Pandas, NumPy, Dash, SciPy, NLTK, Matplotlib, Clustering, TensorFlow, Keras, predictive modelling, Probability & Statistics, Mathematics

Database: MySQL

Deployment: AWS, Sagemaker

Software/Tools: Docker, Kubernetes , OpenVino , Github, Terraform

Education

Bachelor's of Technology in Computer Science (CGPA : 8.4)	2017 - 2021
Jamia Hamdard , New Delhi , India	